

- 4** A particle of mass 0.5 kg moves along a horizontal straight line. Its velocity is $v\text{ ms}^{-1}$ at time $t\text{ s}$. The forces acting on the particle are a driving force of magnitude 50 N and a resistance of magnitude $2v^2\text{ N}$. The initial velocity of the particle is 3 ms^{-1} .

(a) Find an expression for v in terms of t . [7]

[illegible]

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(b) Deduce the limiting value of v .

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